

SUSHANT LOKHANDE

San Jose, CA | +1 669-340-6655 | lokhandesushant014@gmail.com | sushantlokhande.me |
linkedin.com/in/sushantlokhande14 | github.com/sushantlokhande14

EXPERIENCE

San Jose State University | Graduate Research Assistant, Systems Testing & Evaluation Aug 2025 to May 2026

- Validated **14 interoperability scenarios** across networking, dependency, and **failure-recovery** paths to confirm recovery behavior before large experiment runs.
- Built an automated evaluation harness for controlled experiments, cutting setup effort **55%** and making every result reproducible from a single config.
- Reduced rebuffering **19%** versus a fixed-rate baseline by building a **Docker**, **WebTransport**, and **Media-over-QUIC** testbed and running controlled adaptation experiments across network and protocol configs.
- Boosted malware classification to **80.2%** across 17 families with a soft-voting **PyTorch** ensemble (CNN, ResNet50, ViT), **2.4 pts** over the best single model at $p < 0.001$.

Turito | Software Engineering Intern, Reliability & Performance Jun 2022 to Oct 2022

- Cut controlled-disruption failures **42%** (3.1% to 1.8%) by hardening **retries**, timeouts, failure recovery, and distributed state transitions.
- Reduced production diagnosis time **35%** by adding structured **metrics**, runtime-state visibility, and targeted logs.
- Slashed p95 **CPU 22%** by profiling high-frequency media paths and eliminating excess buffer allocation, payload copies, repeated routing lookups, and **lock contention**.
- Debugged **ICE/STUN/TURN** connection setup and degraded-network recovery across a **WebRTC** control plane and **RTP/RTCP** media paths spanning about **12 regions**.

Trivia Softwares | Software Engineering Intern Nov 2021 to Jan 2022

- Built a **Rust/Tokio** load-testing framework driving about **25,000** concurrent sessions, exposing lock contention, serialized **WebSocket** writes, and slow session cleanup.
- Cut performance-investigation time about **40%** by building tooling on Rust workers, **Python** aggregation, **SQLite**, Flask, and Chart.js.
- Raised equivalent-load signaling throughput about **31%** by surfacing concurrency bottlenecks that drove targeted backend fixes.

TECHNICAL SKILLS

Languages: Python, Java, C++17, TypeScript, SQL, Rust

Testing & QA: pytest, Selenium, Locust load testing, regression suites, failure classification, interoperability scenarios, CI/CD

Automation & Tooling: test harness design, telemetry pipelines, structured metrics and logging, Docker, Git, Linux

Systems Under Test: REST and gRPC services, WebRTC and RTP/RTCP media paths, QUIC, Kafka, PostgreSQL, Redis

PROJECTS

Teller, Agent Evaluation Environment | *Browser agents, Claude Code, Python, TypeScript*

- Analyzed **14** computer-use agent runs against an adversarial app (nested frames, missing test IDs, session expiry), classifying each failure as planning, execution, validation, or recovery.

Document Action Agent | *React, TypeScript, FastAPI, LLMs*

- Built a document-grounded assistant that verifies sources before recommending actions and routes ambiguous cases to **human review** instead of completing them silently.

TCP Sliding Window Simulator | *Python, raw sockets, Linux*

- Implemented handshake, sliding window, cumulative ACKs, and retransmission over raw sockets with live packet-state visualization.

OPEN SOURCE

- microsoft/markitdown**: fixed EPUB conversion for percent-encoded manifest paths, with regression tests (merged)
- fission/fission**: 3 merged PRs, Kubernetes event and KEDA object capture in support dumps, removed duplicate package enumeration

EDUCATION

San Jose State University | M.S. in Computer Science | GPA 3.8/4.0 | Spartan Scholar Aug 2024 to May 2026

Coursework: Machine Learning, Distributed Computing, Database Systems, Computer Networks, Advanced Algorithms, Artificial Intelligence

University of Mumbai | B.Tech. in Computer Engineering | GPA 3.9/4.0 Jun 2019 to May 2023

Research: malware image-classification ensemble selected for Dr. Mark Stamp's ML textbook; sole-author contactless palmprint verification (0.9975 AUC)